

Paper Session 5: Digital Health

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CITATION AND MERITORIOUS AWARD WINNER

DEFINING OVEREATING PHENOTYPES IN NATURALISTIC SETTINGS: LEVERAGING MOBILE HEALTH AND MACHINE LEARNING

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Despite the known complexity of overeating, prior research largely examines single causal mechanism of overeating (e.g., stress, loss of control). With recent advancements in mobile and wearable computing, we can examine—in natural settings—multiple behavioral, environmental, and psychological factors and how they co-occur with each other. This allows us to identify behavioral phenotypes warranting different treatments. In our SenseWhy study, we use wearable sensing and EMA to record rich, multi-modal data from thousands of eating episodes and identify phenotype clusters associated with overeating behaviors.

During waking hours, 48 participants wore a wearable camera, 2 passive sensing devices, and a food-tracking app for 2 weeks. We define overeating as any meal exceeding the 1 standard deviation of a participant's mean calorie intake per meal. We employed a semi-supervised learning clustering pipeline by leveraging a deep neural net and UMAP nonlinear manifold learner to eliminate irrelevant features and improve visual interpretability. We then utilized K-means clustering followed by Z-score analysis on the separable clusters to pinpoint leading factors driving problematic eating behaviors. We applied these ML approaches separately to EMA-only data, passive-sensing only data, and the combined EMA and passive-sensing data.

Our EMA-only, passive-sensing only, and combined approach yielded six, three, and three distinct clusters within overeating meals. Within our most comprehensive analysis (i.e., EMA-only), out of 2246 recorded meals (369 (16.4%) overeating meals), we identified six distinct overeating phenotypes, each reported here with a descriptive name, cluster purity (%), count of overeating meals (n), and concise interpretation:

“Stress-driven Night Nibbling”: (48%, 27), eating while stressed or lonely at night

“Night Craving”: (86%, 113), heightened biological hunger at night and opting for self-prepared meals

“Uncontrolled Pleasure Eating”: (78%, 39), hedonic and loss-of-control eating during task-oriented distractions (e.g., working/studying)

“Nighttime Restaurant Reveling”: (95%, 61), pleasurable social dining during nights

“Take-out Feasting”: (100%, 71), indulgent take-out meals in social settings

“Unstressed Home Dining”: (50%, 4), low-stress meals during task-oriented distractions (e.g., watching TV)

The identified phenotypes represent a unique approach to understanding problematic eating behavior in naturalistic settings, identifying specific combinations of multi-level factors that are likely to co-occur within overeating episodes. This methodology has the potential to support populations struggling with obesity by fostering a more comprehensive understanding of their eating patterns, paving the way for personalized treatment strategies and empowering them to better regulate their eating behaviors.

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IDENTIFY PREDICTORS OF OVEREATING FROM REAL-TIME WEARABLE SENSORS AND ECOLOGICAL MOMENTARY ASSESSMENT IN PEOPLE WITH OBESITY

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Background: Dietary intake often extends beyond calorie and nutrient intake, which are not often captured by current dietary self-reported assessments. Advances in mobile computing can improve the validity of diet assessments using objective measurement and enable real-time capture of factors that surround one's eating behavior. Using wearable sensing and event-based ecological momentary assessment (EMA), we build machine-learned (ML) models using behavioral, social, and psychological features to assess which are most predictive of overeating in people with obesity. Our results aim to empower our ability to design timely interventions to detect, predict, and preempt overeating.

Methods: 60 participants with obesity (BMI ≥ 30 kg/m²) completed event-based EMAs and wore a smartwatch, necklace sensor, and activity-oriented camera for 14 waking days. The EMAs captured social and psychological features, and foods consumed during each eating episode. To verify food records, participants completed daily 24-hour diet recalls. The sensor devices continuously captured video and physiological metrics. Overeating was defined as any meal exceeding 1 standard deviation of a participant's mean per-meal calorie intake. We build ML models using XGBoost (a highly efficient, flexible, and portable algorithm) based on features captured from sensors (sensor only), features estimated from EMA (EMA only), and features captured from both (sensor+EMA) to classify meals as overeating or non-overeating.

Results: The EMA only dataset (n=48, 2,302 meals, 368 overeating) achieved an AUC score of 0.814, the 5 most predictive features being *food type*, *food source*, *night eating*, *eating in absence of hunger (EAH)*, and *pre-meal biological hunger*. The sensor only dataset (n=41,700 meals, 109 overeating) resulted in an AUC score of 0.732, and the 5 most predictive features were *meal duration*, *# of bites*, *inter-chew interval*, and *eating speed*. The combined sensor+EMA dataset resulted in an AUC score of 0.826, and the 5 most predictive features were *# of bites*, *pre-meal biological hunger*, *food source*, *post-meal loss of control (LOC)*, and *pre-meal loneliness*.

Conclusions: We report on a novel SenseWhy system that predicts overeating from EMA, passive sensor, and combined sensor+EMA datasets. Features identified across all datasets may guide future studies of the contexts and behavioral aspects of eating and overeating episodes. Results from our analysis of sensor-only data suggests the possibility of wearable systems capable of predicting and intervening on overeating episodes in real-time without any active input from the user.

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